



Fraud Detection in Telecommunication Companies

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Abstract

Telecommunication fraud results in significant financial losses and affects customer trust. This study explores fraud detection methods using historical call data records (CDRs), user behavior analytics, and additional datasets. Machine learning techniques such as Logistic Regression, XGBoost, and Neural Networks were applied. XGBoost was the best-performing model with an accuracy of 90.87%, precision of 94.21%, recall of 86.96%, and an AUC score of 0.91. The research demonstrates the feasibility of deploying AI-driven fraud detection systems in real-time.

Problem Statement

Telecommunication fraud results in billions of dollars in losses annually, impacting 2.5% of total revenues and eroding customer trust. Traditional rule-based detection methods struggle to adapt to evolving fraud tactics, leading to high false-positive rates and delayed identification of fraudulent activities. Fraudsters exploit technological vulnerabilities, regulatory loopholes, and social engineering tactics to bypass security measures. Given the high volume of transactions, there is an urgent need for real-time fraud detection using advanced machine learning (ML) models to enhance accuracy, reduce false positives, and improve operational efficiency in fraud prevention.

Basic Concepts

Telecom Fraud Detection: Fraud detection relies on analyzing historical Call Data Records (CDRs), user behaviour patterns, and additional datasets (SMS, Internet Usage, and User Information).

Machine Learning Approach: Advanced ML models such as Logistic Regression, XGBoost, and Neural Networks are used to classify fraudulent vs. legitimate transactions.

Feature Engineering: Key fraud indicators include call duration, frequency, transaction patterns, and anomalies in user behaviour.

Data Preprocessing: Techniques such as oversampling (SMOTE), feature selection, and hyperparameter tuning are implemented to handle imbalanced data and improve model performance.

Methodology

Data from Sichuan Telecom (August 2019 - March 2020) was analyzed. The dataset contained four sources:

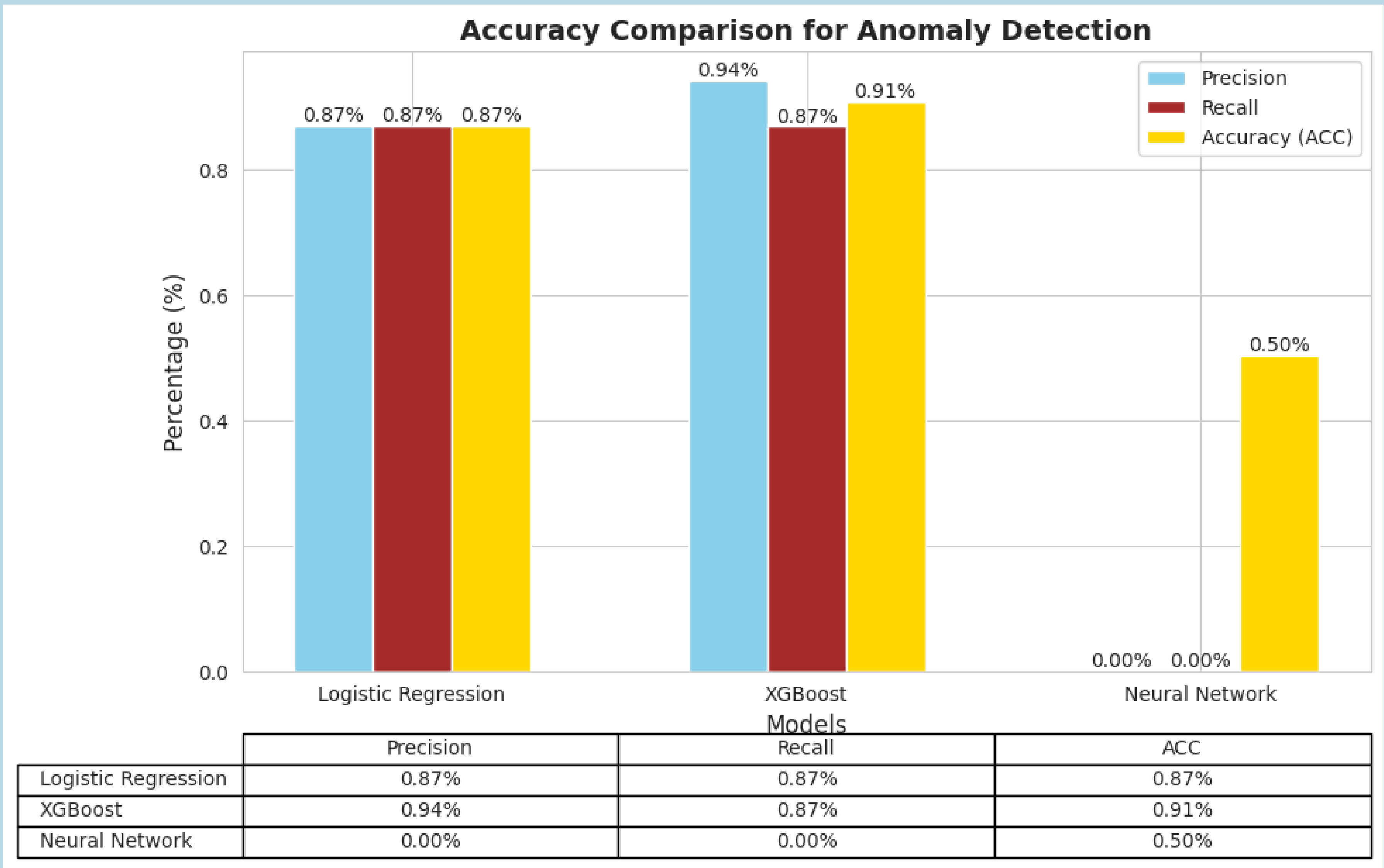
- Call Data (VOC)
- SMS Records
- User Information (USER)
- Internet Usage Data (APP)

The models used were Logistic Regression, XGBoost, and Neural Networks. Feature engineering, hyperparameter tuning, and oversampling techniques were applied to improve model performance.

Results

Model Performance Comparison:

Model	Accuracy	Precision	Recall	AUC
Logistic Regression	87%	87%	87%	0.87
XGBoost	91%	94%	87%	0.91
Neural Networks	50%	0%	0%	0.50



figureModel Performance Metrics

Key Findings:

- XGBoost was the best-performing model.
- Logistic Regression provided a stable baseline.
- Neural Networks struggled due to data scarcity.

Conclusion

Machine learning significantly enhances fraud detection capabilities in the telecom industry. The study demonstrated that XGBoost provides optimal accuracy and precision, making it a viable choice for deployment in fraud detection systems. Further research can explore hybrid models and real-time adaptation techniques to counter evolving fraud tactics.

References

References

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